ISSN 1831-9424

European Commission

Review of Risk Assessment Models for Breast Cancer

Morgano, G.P., Garcia Escribano, M., Parmelli, E., Riggi, E., Janusch Roi, A., Broeders, M., Castells, X., Fitzpatrick, P., Follmann, M., Giordano, L., Graewingholt, A., Hofvind, S., Nystrom, L., Quinn, C., Schunemann, H., Amato, L., Vecchi, S., De Crescenzo, F., Alonso Coello, P., Rigau Comas, D., Canelo-Aybar, C., Nieto-Gutierrez, W.

2024

ECIBC

EUROPEAN COMMISSION INITIATIVE ON BREAST CANCER

Joint Research Centre

EUR 40013

Contact information
Name: European Commission, Joint Research Centre, Disease Prevention Unit
Address: Via Enrico Fermi 2749, TP 127, 21027 Ispra (VA), Italy
Email: JRC-CANCER-POLICY-SUPPORT@ec.europa.eu

EU Science Hub
https://joint-research-centre.ec.europa.eu

JRC138682

EUR 40013

Print	ISBN 978-92-68-20576-1	ISSN 1018-5593	doi:10.2760/2311237	KJ-01-24-001-EN-C
PDF	ISBN 978-92-68-20459-7	ISSN 1831-9424	doi:10.2760/0359784	KJ-01-24-001-EN-N

Luxembourg: Publications Office of the European Union, 2024

How to cite this report: European Commission, Joint Research Centre, Morgano, G.P., Garcia Escribano, M., Parmelli, E., Riggi, E., Janusch Roi, A., Broeders, M., Castells, X., Fitzpatrick, P., Follmann, M., Giordano, L., Graewingholt, A., Hofvind, S., Nystrom, L., Quinn, C., Schunemann, H., Amato, L., Vecchi, S., De Crescenzo, F., Alonso Coello, P., Rigau Comas, D., Canelo-Aybar, C. and Nieto-Gutierrez, W., *Review of Risk Assessment Models for Breast Cancer*, Publications Office of the European Union, Luxembourg, 2024, https://data.europa.eu/doi/10.2760/0359784, JRC138682.

CONTENTS

Abstract 5

Acknowledgements 6

1 Introduction 7

2 Methods 8

2.1 Overview of reviews assessing RAMs 8

2.1.1 Eligibility criteria 8

2.1.2 Search strategy 8

2.1.3 Study selection and data extraction 8

2.1.4 Quality of reviews and risk of bias of RAMs 8

2.1.5 Synthesis of results 9

2.2 Review of deep learning prediction models 9

2.2.1 Eligibility criteria 9

2.2.2 Search strategy 10

2.2.3 Data extraction and synthesis of results 10

3 Results 11

3.1 Overview of reviews 11

3.1.1 Selection and characteristics of reviews 11

3.1.2 Overlapping and quality of reviews 12

3.1.3 Characteristics and development of RAMs 12

3.1.4 Models validation 13

3.1.5 Risk of bias of RAMs 14

3.2 Primary studies of deep learning prediction models 14

3.3 Ongoing studies 14

4 Conclusion 15

References 16

List of abbreviations 18

List of figures 19

List of tables 20

Annexes 21

Annex 1. Search strategy 21

Annex 2. Excluded studies 26

Annex 3. Tables 27

ABSTRACT

Background

Breast cancer is one of the most common cancers affecting women worldwide, with the number of new cases predicted to increase to over three million and the number of deaths to one million per year, by 2040. To prevent breast cancer, there is now increasing demand for personalisation of breast cancer screening based on risk levels, considering the individual characteristics of the women.

Methods

We performed a synthesis of the evidence available for risk assessment models (RAMs) that aimed to predict the short and long-term risk of being detected with breast cancer. We followed the Cochrane Handbook for Systematic Reviews of Interventions, and adhered to the Preferred Reporting Items for Overviews of Reviews (PRIOR) statement for the reporting.

Results

We identified nine systematic reviews that evaluated risk assessment models for breast cancer. Across the systematic reviews, we identified five original models and 13 modifications that were externally validated. The Gail or BCRAT model was the most frequently evaluated, followed by the Rosner and Colditz model.

Conclusions

The results from the external validation should be taken with caution due to the probability of high risk of bias of the included studies. The evidence from primary studies that include deep learning prediction models is limited. Most of the studies only report internal validation results in samples with small number of events.

ACKNOWLEDGEMENTS

We thank Nancy Santesso (chair of the European Commission Initiative on Breast Cancer's Guideline Development Group) for her review and comments to this evidence synthesis effort.

Authors

European Commission's Joint Research Centre:

Gian Paolo MORGANO, Marta GARCIA ESCRIBANO, Elena PARMELLI, Emilia RIGGI, Annett JANUSCH-ROI.

The European Commission Initiative on Breast Cancer's Guideline Development Group, PICO Responsible Unit:

Mireille BROEDERS, Xavier CASTELLS, Patricia FITZPATRICK, Markus FOLLMANN, Livia GIORDANO, Axel GRAEWINGHOLT, Lennarth NYSTRÖM, Cecily QUINN, Holger SCHÜNEMANN.

Department of Epidemiology Lazio Region - ASL Rome:

Laura AMATO, Simona VECCHI, Franco DE CRESCENZO.

Within the contract F.B798738.

Asociación Colaboración Cochrane Iberoamericana:

Pablo ALONSO COELLO, David RIGAU COMAS, Carlos CANELO AYBAR, Wendy NIETO GUTIERREZ.

Within the Framework Contract 943473.

1 INTRODUCTION

Breast cancer is one of the most common cancers affecting women worldwide, with the number of new cases predicted to increase to over three million and the number of deaths to one million per year, by 2040 [1]. To prevent breast cancer, there is now increasing demand for personalization of breast cancer screening based on risk levels, considering the individual characteristics of women [2, 3].

Over the last few decades, several risk factors for breast cancer have been identified, such as age, ethnicity, younger age at menarche, parity, older age at first full-term pregnancy, exogenous hormones, breast density, family history of cancer, benign lesions, and genetic traits [4]. However, previous studies have shown that individual predictors have limited explanatory power [5]. To address this issue, it has been suggested that multiple predictors risk-assessment models (RAMs) could integrate minor effects and predict better the likelihood of breast cancer risk [5]. As a result, several risk prediction models have been developed to estimate the probability of breast cancer risk, within a specified time period [2].

The ability to predict the risk of being detected with breast cancer in the short term (e.g. 2 years or less) would allow proposing personalised strategies after each round of screening, while predicting the long term risk (e.g. 5 years or more) would allow proposing more general screening strategies based on risk levels. However, the effectiveness of using RAMs to personalise breast cancer screening based on risk levels is yet to be established, and randomized controlled trials are currently ongoing [6].

We have conducted this review as part of the 2021 update of the European breast cancer guidelines on screening and diagnosis with the aim of summarising the existing RAMs for prediction of risk for breast cancer in asymptomatic women [7].

2 METHODS

We performed a synthesis of the evidence available for risk assessment models (RAMs) to predict breast cancer risk in two phases: a) an overview of reviews of RAMs, and b) a review of deep learning models. The methods for both phases are described below.

2.1 OVERVIEW OF REVIEWS ASSESSING RAMS

We conducted an overview of reviews assessing RAMs that aimed to predict the short and long-term risk of being detected with breast cancer. We followed the Cochrane Handbook for Systematic Reviews of Interventions [8], and adhered to the Preferred Reporting Items for Overviews of Reviews (PRIOR) statement for the reporting [9].

2.1.1 Eligibility criteria

We included systematic reviews with a clearly structured research question, applying systematic and explicit methods to identify and synthesize primary studies, and reporting the development or validation of original RAMs to predict the risk of being detected with breast cancer in women in the general population.

The eligible reviews included studies that reported: any type of modelling method strategy (i.e. including deep learning); used data from cross-sectional studies, cohort studies, case–control studies and randomised controlled trials; were developed for the population of asymptomatic women, and; reported a multivariable (i.e., at least two variables or predictors) model. We excluded narrative review articles, editorials or letters, comments, and conference proceedings.

2.1.2 Search strategy

An initial comprehensive search was conducted in MEDLINE (Ovid 1946 to May 2021), Embase (Ovid 1974 to May 2021), Cochrane Library (all years to May 2021), Epistemonikos, and Web of Science (from inception until May 2021). An information specialist developed the search strategy in MEDLINE, and then it was adapted for use in the other databases. The full search strategy for MEDLINE is available in Annex 1. The literature search was not restricted by language or date of publication. To expand our search, reference lists of the retrieved articles were also screened.

2.1.3 Study selection and data extraction

Two researchers independently reviewed the titles and the abstracts of the retrieved articles, applying the inclusion and exclusion criteria mentioned above. The same two researchers independently reviewed the full-text version of the articles, to confirm their eligibility for inclusion. Disagreements were resolved in a consensus meeting. We report in Annex 2 the reviews that were excluded and the reasons for exclusion.

From the included reviews, we collected information regarding the publication (author names, journal, year of publication, and country of origin), databases searched, and methods reported. Then, for each RAM identified by the reviews, we extracted the characteristics of the target population, setting, risk factors included time frame of prognostic measurement, outcomes included (e.g. in situ or invasive breast cancer), calibration, discrimination, and type of validation.

2.1.4 Quality of reviews and risk of bias of RAMs

We assessed the risk of bias of the reviews using the Risk of Bias in Systematic Reviews (ROBIS) tool [10]. We focused on four domains: (a) study eligibility criteria, (b) identification and selection of studies, (c) data collection and study appraisal and (d) synthesis and findings. We provide the critical appraisal of the individual RAMs, based on the risk-of-bias tool and the judgments reported in each review. If multiple systematic reviews provided an assessment for the same RAM, we relied on the systematic reviews that reported the use of the Prediction model Risk Of Bias ASsessment Tool (PROBAST), a tool for assessing the risk of bias and applicability of diagnostic and prognostic prediction model studies [11].

2.1.5 Synthesis of results

We assessed the overlap of included studies across systematic reviews. In case of important overlap, we prioritized the results from the reviews that covered the largest segment of the body of evidence, were at low risk of bias, and provided the most updated search. We narratively summarized the results from each systematic review through tabulated summaries. If no overlap was identified, we provided the results regardless of the characteristics of the review.

We present the estimates for discrimination and calibration for each model derived from the validation studies, prioritizing the estimates from those RAMs that were externally validated (Table 1). For RAMs that were modifications of previously reported models, we only included those that were evaluated in more than two studies, and reported information on performance.

Table 1. Definitions of the parameters used to evaluate the performance of the risk assessment models for breast cancer.

Parameters	Definition
Discrimination	Measures how well a model can differentiate (discriminate) between high and low-risk women, evaluated through AUROCC (Area under the receiver operating characteristic curve) and c-statistics values. These measures are reported in a range from 0 to 1 (perfect discrimination). We consider that an appropriate risk model must be at least better than 0.70 (at least better than chance or clinical judgment); higher values indicate better performance.
Calibration	Measures how well the model-predicted risk estimates correspond to absolute risks observed in validation datasets, comparing the average predicted breast cancer risk with the overall observed breast cancer rate (E/O-Expected/Observed ratio or O/E-Observed/Expected ratio). A mismatch indicates that on average the model over - or underestimates the risk, on average.
Internal validation	The original prediction model was tested in the sample in which it was constructed, to determine whether the model works to a satisfactory degree.
External validation	The original prediction model was tested on a set of women/screening participants different from those on whom it was originally constructed, to determine whether the model works to a satisfactory degree.

2.2 REVIEW OF DEEP LEARNING PREDICTION MODELS

Given that our systematic search did not identify any reviews that included models constructed with a machine-learning strategy, we conducted a review of primary studies to assess the performance of deep learning models to assess the risk of being detected with breast cancer.

2.2.1 Eligibility criteria

We included primary studies reporting the development or validations of original deep learning models to predict the risk of being detected with breast cancer. The eligible models were developed for asymptomatic women, and reported a multivariable (i.e., at least two variables or predictors) model.

2.2.2 Search strategy

We did not conduct a systematic search from biomedical databases. We considered the date of search from the last review included in the overview, to conduct the search for primary studies. We also requested the multidisciplinary experts from the ECIBC's Guidelines Development Group to provide any recent relevant publications that fulfilled our eligibility criteria.

2.2.3 Data extraction and synthesis of results

We collected information regarding the publication (author names, year of publication, and country of origin), characteristics of the model, and the performance (discrimination and calibration) derived from the model.

We provide a narrative summary of the characteristics of the primary studies, including information on the target population, time horizon, predictors, and performance estimates. For each model, we report the estimates for discrimination and calibration narratively.

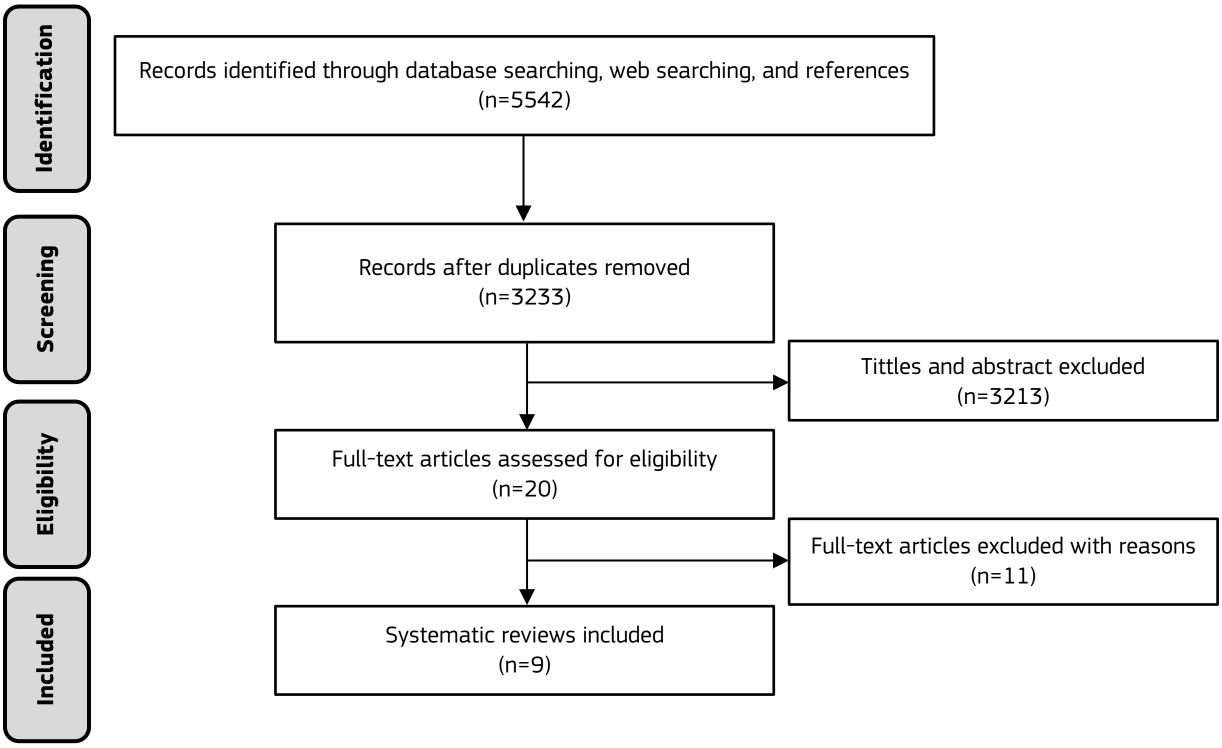
3 RESULTS

3.1 OVERVIEW OF REVIEWS

3.1.1 Selection and characteristics of reviews

From our systematic search, we identified 3 233 citations after removing duplicates. We initially selected 20 systematic reviews after reviewing the title and abstract and finally included nine after assessment of the full text [12-20] (Figure 1). A list of excluded systematic reviews with reasons for exclusion is available in Annex 2.

Figure 1. PRISMA flowchart



The included reviews searched the literature in a range of two to five databases, with the date of search ranging from 2010 to 2019. All of the reviews mainly focused on estimating the risk of being detected with breast cancer in the general population; however, four reviews also included studies that evaluated women with other gynaecological cancers [16, 18, 20], and two reviews included studies on women at higher risk (i.e. family history of cancer) [19, 20]. Most of the systematic reviews provided a narrative synthesis, with three conducting a quantitative synthesis, through a meta-analysis (Table 2 in Annex 3).

The systematic reviews included between 12 and 27 primary studies (Table 3 in Annex 3). None of the reviews included models based on deep learning for breast cancer risk assessment. The reviews identified five main models (including their modified versions), those most extensively reported and evaluated across different settings: Gail/BCRAT (Breast Cancer Risk Assessment Tool), Rosner and Colditz, Tyrer-Cuzick/IBIS (International Breast Cancer Intervention Study), Lee, and BOADICEA (Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm). Several other minor models were also described across reviews [13-15, 19] : however, these models were either originally derived from previous models but only internally validated on small samples, or singles studies; the reviews did not provide a detailed description of the characteristics or performance of this models (Table 2 in Annex 3).

3.1.2 Overlapping and quality of reviews

We identified 109 primary studies that evaluated at least one RAM for breast cancer across the included reviews. However, the reviews included only a subset of these studies, ranging from 12 [19] to 27 [20], representing approximately 6% to 14% of all studies. This may be due to differences in inclusion and exclusion criteria, the date of the search, and the number of databases. None of the primary studies were included in all reviews, but five studies were consistently included in most of them (n=5). The overlap of primary studies across the reviews is described in Table 3 (Annex 3).

The risk of bias assessment of the systematic reviews included is detailed in Table 4. Of the nine reviews included, six [12-14, 17-19] were considered to be at high risk of bias and three were considered to be at low risk [15, 16, 20]. The domain with the highest risk of bias was 'data collection and study appraisal', followed by 'synthesis and findings'. For this review, we prioritized the information reported in Meads 2011 [16], Wang 2018 [20], and Louro 2019 [15], whenever possible, as they included the largest number of primary studies and have the lowest risk of bias.

Table 4. Risk of bias assessment of the systematic reviews.

Study	Domains				
	Study eligibility criteria	Identification and selection of studies	Data collection and study appraisal	Synthesis and findings	Risk of bias in the review
Anothaisintawee 2011	High	High	High	High	High
Al-Ajmi 2018	High	High	High	High	High
Fung 2018	Low	Low	High	Low	High
Louro 2019	Low	Low	Low	Low	Low
Meads 2011	Low	Low	Low	Low	Low
Solikhah 2019	High	High	High	High	High
Stegeman 2012	High	Low	High	Low	High
Vilmun 2020	Unclear	Low	High	High	High
Wang 2018	Low	Unclear	Low	High	Low

3.1.3 Characteristics and development of RAMs

The characteristics of the five main prediction models identified are described in Table 5 (Annex 3). Among them, the most frequently assessed model was the Gail/BRCAT model, developed from an American women database to predict the five-year or lifetime risk for in situ and invasive breast cancer, in women from the general population. It incorporates the variables age, reproductive history, and demographic characteristics, as predictive factors. This model has been adapted several times to include ethnic groups (Caucasian-American, African-American, and Asian-American), genetic factors (single nucleotide polymorphism), mammographic characteristics (breast density), nipple aspirate fluid cytology, and other predictors.

The modifications validated and applied in more than two studies are: Gail 2, Gail 3, and BCSC models. Gail 2 and 3 are specifically developed to predict the risk of invasive breast cancer in Caucasian-American (Gail 2), African-American, and Asian-American (Gail 3) general population. The BCSC model includes family history, age, mammographic density, and demographic factors, and was developed to predict the five years-risk of in situ and invasive breast cancer in the general population and postmenopausal women.

The Rosner and Colditz model was developed from a database of the Nurses' Health Study for the prediction of invasive breast cancer. The model includes factors such as lifestyle, hormonal, reproductive, and demographic characteristics (Table 8 in Annex 3). We identified one modification of the Rosner and Colditz model that added predictors, such as the type of benign breast disease, type of menopause, hormone replacement therapy, weight, height, and alcohol; however, the reviews did not provide information neither on the time horizon nor the calibration results.

The Lee model was developed as a combination of the Gail and the Rosner and Colditz models, to estimate the risk of invasive breast cancer in the general population.

The Tyrer-Cuzick or IBIS model includes family history, age, hormonal, reproductive, and demographic factors. There are two modifications of this model, one including breast density and the other including single nucleotide polymorphism as predictors; however, only the latter reported estimates of performance from external validation.

Finally, the BOADICEA model includes many predictors such as family history, sex, age, genetics, lifestyle, hormonal, reproductive, mammographic density, breast tumour pathology, and demographic factors.

3.1.4 Models validation

The reviews provided estimates of performance from external validation for the Gail (including the adaptation Gail 2, Gail 3, BCSC), Rosner and Colditz, and Tyrer-Cuzick models. The Lee model only underwent internal validation, and the results were reported in Al-Ajm 2018 [12]. We prioritized the estimates form external validation, and provided only a brief summary of internal validation estimates (a detail description of these estimates is presented in Table 9 in Annex 3).

Internal validations from original and derived models

Overall, the reviews did not provide information about internal validation of the development of the original Gail and Rosner and Colditz models. On the other hand, the modified versions of these models (Gail and Rosner and Colditz including SNP or breast density, Gail including nipple aspirate fluid cytology, BCSC, CARE, Boyle, Czech, Gail 2, Colditz, and Rosner and Colditz including type of benign breast disease, type of menopause, hormone replacement therapy, weight, height, and alcohol) did not show good discrimination (with a discrimination AUC or c-statistic less than 0.70); moreover, the calibration estimates of the modified Gail models were heterogeneous (E/O values varied from 0.68 to 1.38), while the modification of the Rosner and Colditz model (Rosner and Colditz included type of benign breast disease, type of menopause, hormone replacement therapy, weight, height, and alcohol) overestimated the risk of breast cancer (E/O 33.22).

The addition of single nucleotide polymorphism as predictor slightly improved the discrimination from the Tyrer-Cuzick (AUC improvement 0.05, CI95% 0.04 - 0.06) and BOADICEA models (AUC improvement 0.04, CI95% 0.03 - 0.05); however, the calibration results for these models are not available. The addition of breast density in the Tyrer-Cuzick model reported a good calibration (O/E 0.98).

External validation performance

The models that had the best estimates for calibration were the BCSC (E/O 0.95 to 1.05) and the Rosner and Colditz model (E/O 1.01, CI95% 0.94 - 1.09); while for discrimination the best estimates come from the Gail (AUC 0.70, CI95% 0.57 - 0.77) and the BCSC (AUC 0.64 to 0.69) models. However, some studies that reported estimations for the Gail model included high-risk populations. The results of external validation for each model are described below.

A total of six reviews reported the external validation of the Gail model; the one from Wang et al included the largest number of studies, the most recent search, and performed a meta-analysis [20]. This review reported that the model had an E/O value of 1.03 (CI95% 0.76 - 1.40; n= 5; I2=NR), but showed good discrimination, with an AUC of 0.70 (CI95% 0.57 - 0.77) (Table 6)

The Gail 2 model [20] showed a calibration value higher than one (E/O of 1.20; CI95% 1.07 - 1.35; n=23; I2=NR) and suboptimal discrimination (C-statistic: 0.59, CI95% 0.57 - 0.62, n=23, I2=NR). The validation studies of the Gail 3 and BCSC models showed and AUC of 0.54 (CI95% 0.50 - 0.59) [20] and a range from 0.64 to 0.69 [15], respectively. However, the calibration results show that Gail 3 model underestimates the risk (E/O 1.68 to 2.96) [20] and the BCSC reports values close to one (E/O 0.95 to 1.05) (Table 6 in Annex 3) [15].

Five reviews described the validation results of the Rosner and Colditz models [12, 13, 15, 16, 18], but only one synthesized the results in a meta-analysis [16]. This meta-analysis showed good calibration results close to one (E/O 1.01; CI95% 0.94 - 1.09; n=2; I2=not reported), but with a discrimination result less than 0.60 (C-statistic 0.57; CI95% 0.55 - 0.59; n=1). Finally, one systematic review [15] reported that the Tyrer-Cuzick model did not show good discrimination (AUC 0.57); however, the performance slightly improved when breast density was added (AUC 0.61; calibration not assessed) (Table 6 in Annex 3).

3.1.5 Risk of bias of RAMs

The quality assessment of studies that evaluated RAMs was reported in only three reviews [15, 16, 20]. However, none of them used the PROBAST tool to evaluate the risk of bias. The reviews report that the studies on the Gail, Rosner and Colditz, and Tyrer-Cuzick models were at high risk of bias; evaluated with the ISPOR-AMCP-NPC questionnaire (Table 6 in Annex 3).

3.2 PRIMARY STUDIES OF DEEP LEARNING PREDICTION MODELS

We identified five cohort studies [21-24] that were conducted in Sweden, United States of America, Australia, Taiwan, and Brazil. All the studies evaluated deep learning models for breast cancer risk assessment based on mammographic images, and most of them predict five years-risk of breast cancer [21, 23-25]. The characteristics of the development process and predictors, are described in Table 7 (Annex 3). Although the models reported promising discrimination results (AUC of 0.62 to 0.79), most of them (4/5 studies) were only evaluated through internal validations with a small number of events, and did not report calibration results. Therefore, the results are not conclusive.

3.3 ONGOING STUDIES

Risk assessment models for breast cancer are continuously studied and the field is evolving rapidly. We identified six ongoing studies including further prospective cohorts on novel predictors and combinations of predictors, such as the PRISMA [26] , the KARMA [27] and the PROCAS study [28] and randomised trials, such as the Tailored Breast Screening Trial [6], the MyPeBS [29] and the Wisdom study [30] to check the value of the use of RAMs in real world. The ECIBC aims to continue monitoring the evidence related to RAMs for breast cancer over the next years to ensure that recommendations are informed by the current best available evidence.

4 CONCLUSION

We identified nine systematic reviews that evaluated risk-assessment models for breast cancer; however, most of the models had important methodological limitations (in study eligibility and selection, quality appraisal of the primary studies, and synthesis of evidence). Across the systematic reviews, we identified five original models and 13 modifications that were externally validated. The Gail or BCRAT model was the most frequently evaluated, followed by the Rosner and Colditz model. Only three modified models reported an external validation or are evaluated in more than two studies. All of these are models based on the Gail model. The modified models based on the original Gail model did not substantially improve the discrimination and calibration performance, even when single nucleotide polymorphism or breast density were added. The results of the discrimination and calibration for the modifications of the Rosner and Colditz and Tyrer-Cuzick models are not conclusive, considering the few studies that reported their performance. The external validation results suggest that the BCSC and the Rosner and Colditz models may have the best calibration, and the BCSC model may have the best discrimination; however, these results should be taken with caution due to the probability of high risk of bias of the included studies. The evidence from primary studies that include deep learning prediction models is limited. Most of the studies only report internal validation results in samples with small number of events.

REFERENCES

1. Arnold, M., et al., *Current and future burden of breast cancer: Global statistics for 2020 and 2040.* Breast, 2022. **66**: p. 15-23.
2. Kim, G. and M. Bahl, *Assessing Risk of Breast Cancer: A Review of Risk Prediction Models.* J Breast Imaging, 2021. **3**(2): p. 144-155.
3. Onega, T., et al., *Breast cancer screening in an era of personalized regimens: a conceptual model and National Cancer Institute initiative for risk-based and preference-based approaches at a population level.* Cancer, 2014. **120**(19): p. 2955-64.
4. Britt, K.L., J. Cuzick, and K.A. Phillips, *Key steps for effective breast cancer prevention.* Nat Rev Cancer, 2020. **20**(8): p. 417-436.
5. Coughlin, S.S., *Epidemiology of Breast Cancer in Women.* Adv Exp Med Biol, 2019. **1152**: p. 9-29.
6. Paci, E., et al., *[Tailored Breast Screening Trial (TBST)].* Epidemiol Prev, 2013. **37**(4-5): p. 317-27.
7. Commission, E. *European Guidelines on Breast Cancer Screening and Diagnosis*. 2024 [date accessed 19/06/2024]; Available from: https://cancer-screening-and-care.jrc.ec.europa.eu/en/ecibc/european-breast-cancer-guidelines.
8. Cumpston, M., et al., *Updated guidance for trusted systematic reviews: a new edition of the Cochrane Handbook for Systematic Reviews of Interventions.* Cochrane Database Syst Rev, 2019. **10**(10): p. Ed000142.
9. Pollock, M., et al., *Preferred Reporting Items for Overviews of Reviews (PRIOR): a protocol for development of a reporting guideline for overviews of reviews of healthcare interventions.* Syst Rev, 2019. **8**(1): p. 335.
10. Whiting, P., et al., *ROBIS: A new tool to assess risk of bias in systematic reviews was developed.* J Clin Epidemiol, 2016. **69**: p. 225-34.
11. Wolff, R.F., et al., *PROBAST: A Tool to Assess the Risk of Bias and Applicability of Prediction Model Studies.* Ann Intern Med, 2019. **170**(1): p. 51-58.
12. Al-Ajmi, K., et al., *Review of non-clinical risk models to aid prevention of breast cancer.* Cancer Causes Control, 2018. **29**(10): p. 967-986.
13. Anothaisintawee, T., et al., *Risk prediction models of breast cancer: a systematic review of model performances.* Breast Cancer Res Treat, 2012. **133**(1): p. 1-10.
14. Fung, S.M., et al., *Performance of Single-Nucleotide Polymorphisms in Breast Cancer Risk Prediction Models: A Systematic Review and Meta-analysis.* Cancer Epidemiol Biomarkers Prev, 2019. **28**(3): p. 506-521.
15. Louro, J., et al., *A systematic review and quality assessment of individualised breast cancer risk prediction models.* Br J Cancer, 2019. **121**(1): p. 76-85.
16. Meads, C., I. Ahmed, and R.D. Riley, *A systematic review of breast cancer incidence risk prediction models with meta-analysis of their performance.* Breast Cancer Res Treat, 2012. **132**(2): p. 365-77.
17. Solikhah, S. and S. Nurdjannah, *Assessment of the risk of developing breast cancer using the Gail model in Asian females: A systematic review.* Heliyon, 2020. **6**(4): p. e03794.
18. Stegeman, I. and P.M. Bossuyt, *Cancer risk models and preselection for screening.* Cancer Epidemiol, 2012. **36**(5): p. 461-9.

19. Vilmun, B.M., et al., *Impact of adding breast density to breast cancer risk models: A systematic review.* Eur J Radiol, 2020. **127**: p. 109019.

20. Wang, X., et al., *Assessment of performance of the Gail model for predicting breast cancer risk: a systematic review and meta-analysis with trial sequential analysis.* Breast Cancer Res, 2018. **20**(1): p. 18.

21. Dembrower, K., et al., *Comparison of a Deep Learning Risk Score and Standard Mammographic Density Score for Breast Cancer Risk Prediction.* Radiology, 2020. **294**(2): p. 265-272.

22. Schmidt, D.F., et al., *Cirrus: An Automated Mammography-Based Measure of Breast Cancer Risk Based on Textural Features.* JNCI Cancer Spectr, 2018. **2**(4): p. pky057.

23. Yala, A., et al., *A Deep Learning Mammography-based Model for Improved Breast Cancer Risk Prediction.* Radiology, 2019. **292**(1): p. 60-66.

24. Yala, A., et al., *Multi-Institutional Validation of a Mammography-Based Breast Cancer Risk Model.* J Clin Oncol, 2022. **40**(16): p. 1732-1740.

25. Eriksson, M., et al., *Identification of Women at High Risk of Breast Cancer Who Need Supplemental Screening.* Radiology, 2020. **297**(2): p. 327-333.

26. *PRISMA studie*. [date accessed 19/06/2024]; Available from: https://www.prisma-studie.nl/.

27. Gabrielson, M., et al., *Cohort Profile: The Karolinska Mammography Project for Risk Prediction of Breast Cancer (KARMA).* Int J Epidemiol, 2017. **46**(6): p. 1740-1741g.

28. *PROCAS and BC-PREDICT*. [date accessed 19/06/2024]; Available from: https://www.mcrc.manchester.ac.uk/impact-case-studies/procas-and-bc-predict/.

29. *MyPeBS (My Personal Breast Screening)*. [date accessed 19/06/2024]; Available from: https://www.mypebs.eu/.

30. *WISDOM study*. [date accessed 19/06/2024]; Available from: https://www.thewisdomstudy.org/.

LIST OF ABBREVIATIONS

ECIBC	European Commission Initiative on Breast Cancer
PRIOR	Preferred Reporting Items for Overviews of Reviews
RAM	Risk Assessment Model
ROBIS	Risk of Bias in Systematic Reviews

LIST OF FIGURES

Figure 1. PRISMA flowchart 11

LIST OF TABLES

Table 1. Definitions of the parameters used to evaluate the performance of the risk assessment models for breast cancer. 9

Table 4. Risk of bias assessment of the systematic reviews. 12

Table 2. Characteristics of the systematic reviews included 27

Table 3. Matrix of included studies per systematic review 32

Table 5. Prediction models for breast cancer reported in the systematic reviews 40

Table 6. External validation of the prediction models selected 42

Table 7. Primary studies of prediction models based on deep learning were collected by the ECIBC's Guidelines Development Group. 43

Table 8. Variables included in the risk models for breast cancer 45

Table 9. Results of internal validation of the prediction models for breast cancer reported in the systematic reviews 47

ANNEXES

Annex 1. Search strategy

Cochrane Database of Systematic Reviews (Issue 5, 2021)

#1 MeSH descriptor: [Breast Neoplasms] explode all trees

#2 (breast near cancer*):ti,ab

#3 (breast near neoplasm*):ti,ab

#4 (breast near carcinoma*):ti,ab

#5 (breast near tumor*):ti,ab

#6 #1 OR #2 OR #3 OR #4 OR #5

#7 MeSH descriptor: [Risk Assessment] explode all trees

#8 MeSH descriptor: [Risk] explode all trees

#9 risk*:ti,ab

#10 #7 OR #8 OR #9

#11 #6 AND #10

#12 (predict* or model* or decision* or identif* or prognos*):ti,ab

#13 #11 AND #12 in Cochrane Reviews, Cochrane Protocols

Database: Ovid MEDLINE(R) ALL (1946 to May 28, 2021)

1 exp Breast Neoplasms/

2 (breast adj6 cancer$).tw.

3 (breast adj6 neoplasm$).tw.

4 (breast adj6 carcinoma$).tw.

5 (breast adj6 tumo?r$).tw.

6 1 or 2 or 3 or 4 or 5

7 exp RISK/

8 risk*.tw.

9 Risk Assessment/

10 7 or 8 or 9

11 Models, Statistical/

12 (predict* or validat* or rule* or scor*).ti.

13 ((predict* or multicomponent or multivariable) adj3 model*).tw.

14 (predict* adj5 (outcome* or risk* or model*)).tw.

15 ((history or variable* or criteria or scor* or characteristic* or finding* or factor* or value*) adj5 (predict* or model* or decision* or identif* or prognos*)).tw.

16 (decision* adj5 (model* or clinical* or logistic model*)).tw.

17 (decision* adj5 (model* or clinical* or logistic model*)).tw.

18 (observ* adj3 (variation or model*)).tw.

19 exp Artificial Intelligence/ or exp Deep learning/

20 exp Natural Language Processing/

21 exp Neural Networks, Computer/

22 exp Support Vector Machine/

23 Deep learning.tw.

24 Artificial Intelligence.tw.

25 Naive Bayes.tw.

26 bayesian learning.tw.

27 Neural network.tw.

28 Neural networks.tw.

29 exp Natural Language Processing/

30 support vector*.tw.

31 random forest*.tw.

32 boosting.tw.

33 deep learning.tw.

34 machine intelligence.tw.

35 computational intelligence.tw.

36 regression model*.tw.

37 risk model*.tw.

38 or/11-37

39 6 and 10 and 38

40 meta-analysis/ or systematic review/ or meta-analysis as topic/ or "meta analysis (topic)"/ or "systematic review (topic)"/ or exp technology assessment, biomedical/

41 ((systematic* adj3 (review* or overview*)) or (methodologic* adj3 (review* or overview*))).ti,ab,kf,kw.

42……((quantitative adj3 (review* or overview* or synthes*)) or (research adj3 (integrati* or overview*))).ti,ab,kf,kw.

43 ((integrative adj3 (review* or overview*)) or (collaborative adj3 (review* or overview*)) or (pool* adj3 analy*)).ti,ab,kf,kw.

44 (data synthes* or data extraction* or data abstraction*).ti,ab,kf,kw.

45 (handsearch* or hand search*).ti,ab,kf,kw.

46 (handsearch* or hand search*).ti,ab,kf,kw.

47 (meta regression* or metaregression*).ti,ab,kf,kw.

48 (meta-analy* or metaanaly* or systematic review* or biomedical technology assessment* or bio-medical technology assessment*).mp,hw.

49 (medline or cochrane or pubmed or medlars or embase or cinahl).ti,ab,hw.

50 (cochrane or (health adj2 technology assessment) or evidence report).jw.

51 (comparative adj3 (efficacy or effectiveness)).ti,ab,kf,kw.

52 (outcomes research or relative effectiveness).ti,ab,kf,kw.

53 ((indirect or indirect treatment or mixed-treatment) adj comparison*).ti,ab,kf,kw.

54 review*.ti.

55 or/40-54

56 39 and 55

Database: Embase (1974 to 2021 May 28)

1 exp breast cancer/

2 (Breast adj6 (cancer* or neoplasm* or carcinoma* or tumo?r*)).tw.

3 1 or 2

4 exp risk/

5 risk*.tw.

6 4 or 5

7 statistical model/

8 (predict* or validat* or rule* or scor*).ti.

9 ((predict* or multicomponent or multivariable) adj3 model*).tw.

10 ((history or variable* or criteria or scor* or characteristic* or finding* or factor* or value*) adj5 (predict* or model* or decision* or identif* or prognos*)).tw.

11 (decision* adj5 (model* or clinical*)).tw.

12 (observ* adj3 (variation or model*)).tw.

13 exp deep learning/

14 exp artificial intelligence/

15 exp natural language processing/

16 exp artificial neural network/

17 exp support vector machine/

18 exp deep learning/

19 exp artificial intelligence/

20 bayesian learning.mp. or exp Bayesian learning/

21 neural network.tw.

22 neural networks.tw.

23 natural language processing.tw.

24 support vector*.tw.

25 random forest*.tw.

26 deep learning.tw.

27 machine intelligence.tw.

28 computational intelligence.mp.

29 computer reasoning.tw.

30 regression model*.tw.

31 risk model*.tw.

32 or/7-31

33 3 and 6 and 32

34 "systematic review"/ or meta analysis/

35 "meta analysis (topic)"/

36 "systematic review (topic)"/

37 biomedical technology assessment/

38 ((systematic* adj3 (review* or overview*)) or (methodologic* adj3 (review* or overview*))).ti,ab.

39 ((quantitative adj3 (review* or overview* or synthes*)) or (research adj3 (integrati* or overview*))).ti,ab.

40 ((integrative adj3 (review* or overview*)) or (collaborative adj3 (review* or overview*)) or (pool* adj3 analy*)).ti,ab.

41 (data synthes* or data extraction* or data abstraction*).ti,ab.

42 (handsearch* or hand search*).ti,ab.

43 (mantel haenszel or peto or der simonian or dersimonian or fixed effect* or latin square*).ti,ab.

44......(met analy* or metanaly* or technology assessment* or HTA or HTAs or technology overview* or technology appraisal*).ti,ab.

45 (meta regression* or metaregression*).ti,ab.

46 (meta-analy* or metaanaly* or systematic review* or biomedical technology assessment* or bio-medical technology assessment*).mp,hw.

47 (medline or cochrane or pubmed or medlars or embase or cinahl).ti,ab.

48 (cochrane or (health adj2 technology assessment) or evidence report).jw.

49 (comparative adj3 (efficacy or effectiveness)).ti,ab.

50 (outcomes research or relative effectiveness).ti,ab.

51 ((indirect or indirect treatment or mixed-treatment) adj comparison*).ti,ab.

52 review*.ti.

53 or/34-52

54 33 and 53

Web of Science

1. TS=(Breast NEAR/6 (cancer* OR neoplasm* OR carcinoma* OR tumo?r*))
2. TS =(Risk*)
3. #2 AND #1
4. TI=(predict* or validat* or rule* or scor*)

5. TS=((predict* or multicomponent or multivariable) NEAR/3 model*)

6. TS=(predict* NEAR/5 (outcome* or risk* or model*))

7. TS=((history or variable* or criteria or scor* or characteristic* or finding* or factor* or value*) NEAR/5 (predict* or model* or decision* or identif* or prognos*))

8. TS=(decision* NEAR/5 (model* or clinical* or "logistic model*"))

9. #8 OR #7 OR #6 OR #5 OR #4

10. TS=(deep learning OR Artificial Intelligence OR Naive Bayes OR bayesian learning OR Neural network$ OR Natural language processing OR support vector machine$ OR random forest$ OR boosting OR deep learning OR machine intelligence OR computational intelligence OR computer reasoning)

11. TS=((history or variable* or criteria or scor* or characteristic* or finding* or factor* or value*) NEAR/5 (predict* or model* or decision* or identif* or prognos*))

12. TS=(decision* NEAR/5 (model* or clinical* or "logistic model*"))

13. #10 OR #9

14. #11 AND #3

15. TI=review*

16. TS= (systematic* NEAR/3 review*) OR TS= (systematic* NEAR/3 bibliographic*) OR TS= (systematic* NEAR/3 literature*) OR TS=(comprehensive* NEAR/3 bibliographic*) OR TS=(information NEAR/2 synthesis) OR TS=(data NEAR/2 synthesis)

17. TI=(meta-analy* OR metaanaly*)

18. #15 OR #14 OR #13

19. #16 AND #12

Epistemonikos

title:(Breast AND (cancer* OR neoplasm* OR carcinoma* OR tumor* OR tumour*)) AND title:(risk*) AND (title:(predict* OR model* OR decision* OR identif*) OR abstract:(predict* OR model* OR decision* OR identif*))

Annex 2. Excluded studies

Study	Reason for exclusion
Amir, E., et al. (2010). "Assessing women at high risk of breast cancer: a review of risk assessment models." Journal of the National Cancer Institute 102(10): 680-691.	Not a systematic review
Batchu S et al.A Review of Applications of Deep learning in Mammography and Future Challenges. Oncology 2021:1-8	Not relevant objective
Kim G et al. Assessing Risk of Breast Cancer: A Review of Risk Prediction Models. Journal of breast imaging 2021;3(2):144	Not a systematic review
Kim Do Yeun et al. Breast Cancer Risk Prediction in Korean Women: Review and Perspectives on Personalized Breast Cancer Screening. Journal of breast cancer 2020;23(4):331	Not a systematic review
Nelson HD et al. Risk Assessment, Genetic Counseling, and Genetic Testing for BRCA-Related Cancer. Agency for Healthcare Research and Quality (US) 2013	Update in Nelson 2019
Nelson, H. D., et al. (2019). "Risk Assessment, Genetic Counseling, and Genetic Testing for BRCA-Related Cancer." JAMA 322(7): 666-685.	Focused on risk of cancer
Siesling S. Clinical risk prediction models for breast cancer: A review of models developed between 2010 and 2018. Cancer Research 2019;79 (4 Supplement 1)	Conference abstract
Roman M et al. Personalized breast cancer screening strategies: A systematic review and quality assessment. PloS one 2019;14(12): e0226352	Not relevant objective and outcomes (QUALYs and ICER outcomes)
Koleva-Kolarova R et al. Simulation models in population breast cancer screening: A systematic review. Breast 2015;24(4): 354-63	Simulation model for breast cancer screening
Wang Y et al. Development of a risk assessment tool for projecting individualized probabilities of developing breast cancer for Chinese women. Tumor Biology 2014;35: 10861-10869	Not a systematic review
Wang Xuexia et al. A review of cancer risk prediction models with genetic variants. Cancer Informatics 2014;13: 19-28	Not a systematic review, focus on other types of cancers

Annex 3. Tables

Table 2. Characteristics of the systematic reviews included

Author	Objective	Inclusion criteria	Main population Age	Date of the search	Databases	No of studies Study design	Synthesis Quality assessment	Models(*) (No of studies)
Anothai-sintawee 2011	To evaluate the development and performance of existing prediction models, which are used to estimate the risk of breast cancer	Incl: — Observational studies — Published in English — Models that considered more than one risk factor — Had the outcome as breast cancer versus non- breast cancer — Applied any regression equation to build up the prediction model — Reported each model's performance	General population NR	Medline: 1949 to October 2010 Embase: 1974 to October 2010	Medline and Embase	25 studies: 18 studies developed or modified the previous prediction models (2 Nested case-control, 6 case-control, and 10 cohort) and 7 studies to validate models (NR)	Narrative synthesis	— BCRAT o Gail (8) — Modified BCRAT (4) — Modified BCRAT (CARE) (1) — Modified BCRAT (Czech) (1) — Rosner and Colditz (2) — Modified Rosner and Colditz (5) — Other models [a] (5)
Meads 2011	To qualitatively summarize the models and the risk factors they contain, and to collate and meta-analysis model performance across studies	Incl: — Studies developing and/or validating a breast cancer risk prediction model for the general female population — Model that included multiple variables, at least one of which was a modifiable risk factor Excl: — Evaluated breast cancer in men — Included women who already had	General population NR	Until June 2010	Cochrane library, Medline, Embase, CAB Abstracts, and PsychINFO	26 studies: 17 studies developed or modified the previous prediction models (NR) and 9 validations (NR)	Narrative synthesis and metanalysis (for BCRAT, Rosner and Colditz, and Modified Rosner and Colditz) Quality assessment	— BCRAT o Gail (6) — Gail 2 [b] (12) — Rosner and Colditz (2) — Modified Rosner and Colditz (3) — Pike (2) — IBIS o Tyrer-Cuzick (1)

Author	Objective	Inclusion criteria	Main population Age	Date of the search	Databases	No of studies Study design	Synthesis Quality assessment	Models(*) (No of studies)
		breast cancer, benign breast pathology, or is considered a high-risk group — Models investigating single risk factors and predicting genetic mutations rather than cancer — Studies published more than 25 years ago						
Stegeman 2012	To assess to what extent models have been developed for calculating cancer risk for breast cancer, cervical cancer, and colon cancer	Incl: — Studies that presented or evaluated multivariable risk models for breast cancer, cervical cancer or colon cancer — Models had to contain variables that were available without additional testing Excl: — Models that contained laboratory measurements	General population 20-79y	Until August 2010	Medline and Embase	13 studies (7 case-control, 1 nested case-control, 2 cross-sectional, and 3 cohort)	Narrative synthesis	— BCRAT o Gail (7) — Gail 2 [b] (2) — Modified BCRAT (CARE) (2) — Rosner and Colditz (2) — Modified Rosner and Colditz (1)
Al-Ajmi 2018	To describe breast cancer risk predicting models that incorporated modifiable risk factors and/or factors that can be self-reported	Incl: — Risk models were retrieved based on any study design, study population or types of risk factors	General population 20-74y	Until July 2016	PubMed, ScienceDirect, and Cochrane database	14 studies (9 case-control and 5 cohort)	Narrative synthesis	— BCRAT o Gail (1) — Modified BCRAT (9) — Rosner and Colditz (2) — Modified Rosner and Colditz (1) — Lee (1)

Author	Objective	Inclusion criteria	Main population Age	Date of the search	Databases	No of studies Study design	Synthesis Quality assessment	Models(*) (No of studies)
Wang 2018	To evaluate the performance of different versions of the Gail model	Incl: — Publications in English and Chinese language — Studies that validated the performance or modified the BCRAT/Gail model	General population 18-84y	1 January 1989 to 31 July 2016	PubMed, Embase, WANFANG, VIP, and China National Knowledge Infrastructure	29 studies (21 cohort, 5 case-control, and 3 cross sectional): 24 studies for calibration, 26 studies for discrimination, and 13 studies for diagnostic accuracy	Meta-analysis Quality assessment A subgroup and sensitivity analysis	— BCRAT o Gail [c] (5) — Gail 2 [b,d] (20) — Gail 3 (3)
Louro 2019	To evaluate and describe the results of the individualized risk models, which are used to estimate the risk of breast cancer in general population	Incl: — Studies those published in English — Studies that reported model to estimate the individualized risk of breast cancer in general population — Models that assessed more than one risk factor and reported the quantitative characteristics of the risk prediction model Excl: — External validation that replicated previous models without adding any additional information such as a new design for collecting the inputs data, modifications on the risk factors or the risk model method	General population 20-84Y	Until February 2018	Medline, Cochrane Library, and Embase	24 studies (NR)	Narrative synthesis Quality assessment	— BCRAT or (9) — BCS (4) — Rosner and Colditz (5) — IBIS o Tyrer-Cuzick (1) — Other models [a] (5)

Author	Objective	Inclusion criteria	Main population Age	Date of the search	Databases	No of studies Study design	Synthesis Quality assessment	Models(*) (No of studies)
Fung 2018	To identify existing SNP-enhanced breast cancer risk prediction models and assess their performance	Incl: — Studies published in English — Empirical studies assessing risk prediction models for breast cancer in women, which reported outcomes using AUC or its equivalent — Studies that compared between SNP-enhanced risk prediction models and traditional risk prediction models Excl: — Genome-wide association studies — Reviews, narratives, prognostic or diagnostic studies, model development studies, and non-risk prediction studies — Studies that included only BRCA mutation carriers	General population 20-74y	Until January 2018	Embase, Scopus, and PubMed	26 studies (case control)	Metanalysis Quality assessment	— BCRAT (6) — Partial BCRAT (13) — IBIS o Tyrer-Cuzick (3) — BCSC (3) — Other models [a] (5)
Solikhah 2019	To evaluate the effectiveness of the application of the risk assessment tool for developing breast cancer, especially in the Asian context	Incl: — Any type of study published in English and accessible in full-text — Studies that assessed breast cancer risk instruments using the GM applied in Asian populations — Studies that provided sufficient data of the performance of the	General population (Asian) 20-69y	Until June 19, 2019	PubMed, MEDLINE, Scopus, Web of Science, and Google Scholar	20 studies (10 cross sectional, 5 case-control, 5 cohort)	Narrative synthesis	— BCRAT o Gail [e] (20)

Author	Objective	Inclusion criteria	Main population Age	Date of the search	Databases	No of studies Study design	Synthesis Quality assessment	Models(*) (No of studies)
		models included Excl: — Proceedings articles, case reports, scientific conference articles, and reviews						
Vilmun 2020	To identify clinical breast cancer risk prediction models reporting outcomes with and without inclusion of breast density	Incl: — Developed or modified existing clinical breast cancer risk prediction models applicable to the general population of women — Models that included breast density as an additional risk factor — Studies published in English and accessible in full-text	General population 40-75y	1 January 2007 to 18 November 2019	PubMed, Embase, Web of Science, and Cochrane Library	12 studies (8 case control, 4 cohort)	Narrative synthesis	— BCRAT o Gail + breast density (4) — IBIS o Tyrer-Cuzick + breast density [f] (4) — Other models [h] (5)

NR: not reported; Breast Cancer Risk Assessment Tool (BCRAT); BCSC: Breast Cancer Surveillance Consortium; IBIS: International Breast Intervention Study

[*] One study can evaluate more than one model

[a] The study did not provide a detailed description of each model included in the analysis or the individual performance of each model

[b] The model was evaluated in oncological and general population

[c] Studies with high-risk populations are included (2 studies)

[d] Studies with high-risk populations are included (6 studies)

[e] The study did not report the type of BCRAT model included (original or any modification).

[f] Studies with high-risk populations are included (1 study)

[h] The study provides only a description of the BOADICEA and BRCAPRO models in this group. We only include the BRODICEA model in our review, considering that BRCAPRO evaluated the predictive factors for carrying a mutation of the BRCA1 and BRCA2 genes

Table 3. Matrix of included studies per systematic review

Studies	Year	Anothaisintawee 2012	Mead 2011	Stegeman 2012	Al-Ajmi 2018	Wang 2018	Louro 2019	Fung 2018	Vilmun 2020	Solikhah 2019
Abdolell	2015								X	
Adams-Campbell	2007			X						
Adams-Campbell	2009			X						
Allman	2015							X		
Al Otaibi	2018									X
Amir	2003		X			X				
Anothasintawee	2013					X				
Arne	2009		X							
Banegas	2017				X		X			
Barlow	2006	X	X				X			
Bener	2017									X
Bernatsky	2004					X				
Bondy	1994	X	X			X				
Boyle	2004	X	X		X	X	X			
Brentnall	2015					X			X	

Studies	Year	Anothaisintawee 2012	Mead 2011	Stegeman 2012	Al-Ajmi 2018	Wang 2018	Louro 2019	Fung 2018	Vilmun 2020	Solikhah 2019
Brentnall	2018								X	
Buron	2013					X				
Burnside	2016							X		
Ceber	2013									X
Challa	2013									X
Chay	2012					X				X
Chebowski	2007	X		X		X				
Chen	2006	X	X				X			
Colditz	2000	X	X	X	X		X			
Colditz	2004	X					X			
Cook	2009		X							
Costantino	1999	X	X	X		X				
Darabi	2012							X	X	
Dai	2012							X		
Dite	2013							X		

Studies	Year	Anothaisintawee 2012	Mead 2011	Stegeman 2012	Al-Ajmi 2018	Wang 2018	Louro 2019	Fung 2018	Vilmun 2020	Solikhah 2019
Dite	2016							X		
Dartois	2015					X				
Decarli	2006	X	X	X		X	X			
Duan	2014					X	X			
Ebil	2015									X
Erikson	2017						X			
Al-Azzawi	2016									X
Gail	1989	X	X	X	X		X			
Gail	2007	X	X	X	X					
Gao	2012									X
Guo	2017							X		
Haberle	2012								X	
Hu	2015					X				
Higginbotham	2012							X		
Hsieh	2017							X		

Studies	Year	Anothaisintawee 2012	Mead 2011	Stegeman 2012	Al-Ajmi 2018	Wang 2018	Louro 2019	Fung 2018	Vilmun 2020	Solikhah 2019
Husing	2012							X		
Jupe	2014							X		
Keller	2015								X	
Kerlikowske	2015						X			
Kaklamani	2011							X		
Khazaee-Pool	2016									X
Lee	2004	X			X				X	
Lee	2014							X		
Lee	2015				X			X		
Maas	2016							X		
Mac Karem	2001			X						
MacInnis	2012					X				
Matsumo	2011				X		X			X
McCarthy	2015					X				
McKian	2009	X								

Studies	Year	Anothaisintawee 2012	Mead 2011	Stegeman 2012	Al-Ajmi 2018	Wang 2018	Louro 2019	Fung 2018	Vilmun 2020	Solikhah 2019
Mealiffe	2010							X		
Min	2014					X				X
Mirghafourvand	2016									X
Mohammadbeigi	2015									X
Novotoy	2006	X	X		X		X			
Olson	2004					X				
Park	2013				X					X
Pator-Barriuso	2013					X				
Pfeiffer	2013				X					
Powell	2014					X				
Rauh	2012								X	
Rockhill	2003	X	X	X						
Rockhill	2001	X	X	X		X				
Rong	2016					X				
Rosner	1996	X	X	X	X		X			

Studies	Year	Anothaisintawee 2012	Mead 2011	Stegeman 2012	Al-Ajmi 2018	Wang 2018	Louro 2019	Fung 2018	Vilmun 2020	Solikhah 2019
Rosner	2008	X	X				X			
Rosner	1994		X		X					
Saikiran	2015								X	
Schonberg	2015					X				
Schonfeld	2010	X	X			X				
Shephrd	2011								X	
Sueta	2012							X		
Shien	2016						X	X		
Shieh	2017							X		
Speigelman	1994		X	X		X				
Syednoori	2012									X
Tamimi	2010	X								
Tarabishy	2011					X				
Thomas	2016									X
Tice	2005	X								

Studies	Year	Anothaisintawee 2012	Mead 2011	Stegeman 2012	Al-Ajmi 2018	Wang 2018	Louro 2019	Fung 2018	Vilmun 2020	Solikhah 2019
Tice	2005	X	X				X			
Tice	2008	X	X			X	X			
Tice	2015						X			
Tyrer	2004		X				X			
Ueda	2003	X			X		X			
Ulusoy	2010	X	X							X
Vachon	2015						X	X		
van Veen	2018							X	X	
Viallon	2009		X							
Wacholder	2010		X					X		
Wang	2014						X			
Warwick	2014								X	
Wen	2016							X		
Wu	2015							X		
Xu	2013							X		

Studies	Year	Anothaisintawee 2012	Mead 2011	Stegeman 2012	Al-Ajmi 2018	Wang 2018	Louro 2019	Fung 2018	Vilmun 2020	Solikhah 2019
Yilmaz	2011									X
Zheng	2010							X		
Zhang	2018						X			X
Zhao	2017									X
Studies included		25	26	13	14	27	24	26	12	20

Table 5. Prediction models for breast cancer reported in the systematic reviews

Model	Population (applied)	Time horizon	Target cancer
Gail model *Breast Cancer Risk Assessment Tool – BCRAT* (https://bcrisktool.cancer.gov/calculator.html)	General population	5 years, lifetime	In situ/invasive
Including SNP [a**]	General population, non-BRCA mutation carriers, and post/pre menopause	NR	In situ/invasive
Including breast density [a**]	General population	5 years	In situ/invasive
Including nipple aspirate fluid cytology	General population	5 years	Invasive
Modified Gail /– BCS *Breast Cancer Surveillance Consortium - BCS* (https://tools.bcsc-scc.org/BC5yearRisk/intro.htm)	General population and post menopause	5 years	In situ/invasive
Modified Gail – CARE model [a]	General population	7.57 years	Invasive
Modified Gail – Matsumo model [a***]	General population	NR	Invasive
Modified Gail – Boyle model [a***]	General population	NR	Invasive
Modified Gail – Czech model [a**]	General population	5 years	Invasive
Gail 2 model (Caucasian-American Gail)	General population	5 years	Invasive
Gail 3 model (African-American and Asian-American)	General population	5 years	Invasive
Rosner and Colditz model *based on the Nurses' Health Study*	General population	NR	Invasive
Modified Rosner and Colditz - Pike model [a**]	General population	NR	NR
Including type of benign breast disease, type of menopause, hormone replacement therapy, weight, height, and alcohol [a]	General population	NR	Invasive
Modified Rosner and Colditz or Colditz and Rosner model [a]	Post menopause	NR	Hormonal defined breast

Model	Population (applied)	Time horizon	Target cancer
			cancer
Tyrer-Cuzick model *International Breast Cancer Intervention Study - IBIS* (https://ems-trials.org/riskevaluator/)	General population	NR	In situ/invasive
Including breast density [a**]	General population	5-10 years	In situ/invasive
Including SNP [a**]	General population	NR	In situ/invasive
Modified Gail and Rosner – Lee model [a***]	General population	NR	Invasive
BOADICEA model[**] *Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm* (http://ccge.medschl.cam.ac.uk/boadicea/; https://www.canrisk.org)	General population	NR	NR

SNP: single nucleotide polymorphism; NR: not reported
[a] There were very few studies that explored modifications of original model, therefore, we were not considered relevant in this review
** No validations are reported
*** No external validations are reported

Table 6. External validation of the prediction models selected

Model	Study selected	Type of result	Calibration	Time horizon	Discrimination	Time horizon	Quality assessment
Gail model *BCRAT*	Wang 2018[1]	Meta-analysis	E/O 1.03 (0.76-1.40) [n=5; I2=NR]*	5 to 9.1y	AUC 0.70 (0.57-0.77) [n=5; I2=NR]*	7.1y	NR
Gail 2 model	Wang 2018	Meta-analysis	E/O 1.20 (1.07-1.35) [n=23; I2=NR]**	5 to 15y (Mean 5y)	AUC 0.59 (0.57-0.61) [n=23; I2=NR]*	5 to 16.2y	NR
Gail 3 model	Wang 2018	Individual results	E/O 2.51 (2.14-2.96), 1.85 (1.68-2.04), and 1.29 (1.11-1.51)***	5 to 10y	AUC 0.54 (0.50-0.59)	5y	NR
BCSC model	Louro 2019	Individual results	General: E/O 0.95 to 1.05 5 years: E/O 0.98 and 1.04 10 years: E/O 0.95 and 1.05	5 and 10y	AUC 0.64 to 0.69	NR	Global high risk of bias [2]
Rosner and Colditz model	Mead 2011	Meta-analysis	E/O 1.01 (0.94-1.09) [n=2; I2=NR]	NR	C-statistic 0.57 (0.55-0.59) [n=1]	NR	NR
Tyrer-Cuzick model *IBIS*	Louro 2019	Individual results	NR	NR	AUC 0.57 increase 0.61 with breast density	NR	Global high risk of bias [2]

* The meta-analysis included estimates based on high-risk population
** The meta-analysis included estimates based on oncological patients
*** From a cohort from Singapore at 5 years, a cohort from Singapore at 10 years, and from a Korean cohort at 5 years follow-up, respectively
[1] Did not clarify which version of Gail's model was evaluated (original or an adaption)
[2] Evaluated with the ISPOR-AMCP-NPC Questionnaire

Table 7. Primary studies of prediction models based on deep learning were collected by the ECIBC's Guidelines Development Group.

Study	Design	Country	Target population	Time horizon	Predictors	Discrimination and calibration derived from the model
Dembrower 2020	Retrospective cohort	Sweden	All women aged 40-74 years (278 with breast cancer) with healthy control subjects	5y	— Deep learning model based on mammographic images — Age and breast density	— AUC 0.65 — AUC 0.67
Yala 2019	Retrospective cohort	USA	Screening mammograms	At least 5y	— Risk factor–only logistic regression — Model — Deep learning model based on mammographic images — Hybrid model that combined the logistic regression and the image-based deep learning model	— AUC 0.62 — AUC 0.67 — AUC 0.70 *Pre-menopausal women:* AUC 0.79 (0.67-0.97) *Postmenopausal women:* AUC 0.70 (0.65-0.75)
Yala 2021	Cohort	USA, Sweden, Taiwan and Brazil	Screening mammograms	5y	MIRAI model	C-statistic 0.76 to 0.81*

Study	Design	Country	Target population	Time horizon	Predictors	Discrimination and calibration derived from the model
Eriksson 2020	Cohort	Sweden	Women visiting mammography screening aged 40–74 years	5y	— Age, body mass index, family history of breast cancer, use of hormone replacement therapy, use of tobacco and alcohol, and menopausal status — Tumour characteristics, immunohistochemistry markers, and clinical- pathologic variables — Microcalcifications, masses — Breast density — PRSs (including 313 single nucleotide polymorphisms)	*Model 1* [mammographic density, masses, age, and microcalcifications]: AUC 0.73 (0.71 - 0.74) *Model 2* [variables from model 1 + lifestyle and familial risk factors]: AUC 0.74 (0.72 - 0.75) *Model 3* [variables from model 2 + PRSs]: AUC 0.77 (0.75 - 0.79)
Schmidt 2018	Cohort	Australia and USA	Screening mammograms of women with breast cancer (cases) and women without a diagnosis of breast cancer (controls)	NR	20 gray-level co-occurrence matrix (GLCM) features, based on the statistical properties of neighbouring pixels	AUC 0.66 (0.64 - 0.69)

* External validation

Table 8. Variables included in the risk models for breast cancer

Risk factor group	Risk factor category	Gail/ BCRAT	BCSC	BOADICEA	IBIS	Rosner and Colditz
Family history	Explicit family history of breast and other cancers		√	√	√	
	Age at onset		√		√	
	Bilateral breast cancer		√		√	
Sex				√		
Age		35-90	35-74	any	19-85	any
Genetic factors				√		
Rare truncating/pathogenic variants	BRCA1, BRCA2, PALB2, CHEK2, ATM			√		
Common genetic variants	Polygenic risk score			√		
Unobserved genetic effects	Residual polygenic component			√		
Lifestyle/hormonal/reproductive	Height			√		√
	Body mass index			√		
	Parity			√	√	
	Age at first birth	√		√	√	√
	Age at menarche	√		√	√	√
	Age at menopause			√	√	√
	Use of oral contraceptive			√		
	Use of hormone replacement therapy			√	√	√
	Alcohol intake			√		√
	Type of menopause					√

Risk factor group	Risk factor category	Gail/ BCRAT	BCSC	BOADICEA	IBIS	Rosner and Colditz
	Oestradiol level					√
Mammographic density			√	√		√
Breast tumour pathology	Oestrogen receptor status			√		
	Progesterone receptor status			√		
	HER2 receptor status			√		
	CK14 status			√		
	CK5/6 status			√		
Demographic factors	Country of origin			√		
	Birth cohort			√		
	Family ethnicity	√	√	√	√	
	Prior/number of breast biopsies	√	√		√	
	Atypical hyperplasia	√	√		√	
	Benign breast disease					√
	Lobular carcinoma in situ (LCIS)	√	√		√	

Table 9. Results of internal validation of the prediction models for breast cancer reported in the systematic reviews

Models	Internal validation		Derived from the model [a]	
	Discrimination	Calibration	Discrimination	Calibration
Gail model			NR	NR
Including SNP	NR	NR	General (7 to 77 SNPs): AUC improvement 0.03 (0.03 - 0.04)[1] Partial BCRAT (5 to 22 SNPs): Improvement 0.05 (0.02 - 0.09)[1] Partial BCRAT + additional factor (2 to 92 SNPs): Improvement 0.04 (0.03 - 0.06)[1]	NR
Including breast density	NR	NR	General: c-statistic 0.68 (0.66 - 0.70)[2] Percent dense area[3]: Custom software 0.65 (0.63 - 0.67); Cumulus 0.60 (0.58 - 0.62) Absolute dense area[3]: Custom software 0.66 (0.64 - 0.68) In situ/invasive[3]: 0.59 (0.57 - 0.61) Invasive3: 0.59 (0.57 - 0.61)	NR
Including nipple aspirate fluid cytology	NR	NR	c-statistic 0.64[2]	NR
Modified Gail o BCS	C-statistic 0.66 (0.65 - 0.66)[2]	E/O 1.03 (0.99 - 1.06)[2]	c-statistic 0.64[2] AUC 0.64 to 0.69[4] Including SNP[1] (76 to 83 SNPs): AUC improvement 0.04 (0.03 - 0.05)	NR
Modified Gail – CARE model	NR	NR	NR	NR
Modified Gail – Matsumo model	AUC 0.61 (0.59 - 0.64)[5]	NR	NR	E/O 1.17 (0.99 - 1.38)[5]
Modified Gail –	0.58[2]	E/O 0.96 (0.75–1.16)	AUC 0.59[5]	

Models	Internal validation		Derived from the model [a]	
	Discrimination	Calibration	Discrimination	Calibration
Boyle model		and E/O 0.92 (0.68–1.16)[5]		
Modified Gail – Czech model	NR	NR	NR	NR
Gail 2 model	NR	NR	AUC 0.58[6]	E/O 0.94 (0.89 - 0.99)[6]
Gail 3 model	NR	NR	NR	NR
Rosner and Colditz model	NR	NR	NR	NR
Modified Rosner and Colditz - Pike model	NR	NR	NR	NR
Including type of benign breast disease, type of menopause, hormone replacement therapy, weight, height, and alcohol	NR	E/O 33.22[2]	c-statistic 0.64[2] ER+/PR+: c-statistic 0.64 (0.63 - 0.66)[2] ER-/PR-: c-statistic 0.61 (0.58 - 0.64)[2]	NR
Modified Rosner and Colditz or Colditz and Rosner model	NR	NR	c-statistic 0.64 (0.63–0.65)[2]	NR
Modified Gail and Rosner - Lee model	AUC 0.62 (0.62 - 0.623)[5] Under 50: 0.61 (0.60 - 0.61)[5] Above 50: 0.64 (0.63 - 0.64)[5]	NR	NR	NR
Tyrer-Cuzick model	NR	NR	C-statistic 0.762 (0.70 - 0.82)[7]	E/O 0.81 (0.62 - 1.08)[7]
Including breast density*	NR	NR	General[1]: AUC 0.62 to 0.64 In situ/invasive[1]: 0.61 (0.59 - 0.63)	General[1]: O/E 0.98 10 years[1]: O/E

Models	Internal validation		Derived from the model [a]	
	Discrimination	Calibration	Discrimination	Calibration
			Invasive[1]: 0.61 (0.58 - 0.63)	1.17
Including SNP (18 to 77 SNPs)	NR	NR	Improvement AUC 0.05 (0.04 - 0.06)	NR
BOADICEA model	NR	NR	AUC 0.70 (0.67 - 0.73)[1] Including SNP[1] (77 SNPs): AUC improvement 0.04 (0.03 - 0.05)	NR

SNP: single nucleotide polymorphism; NR: not reported

Reported in [1] Fung 2018; [2] Anothaisintawee 2011; [3] Vilmun 2020; [4] Louro 2019; [5] Al-Ajmi 2018; [6] Stegeman 2012; and [7] Meads 2011

* We do not report the values of the high-risk population

[a] Estimates from internal validation reported in the primary study that developed the model

Getting in touch with the EU

In person

All over the European Union there are hundreds of Europe Direct centres. You can find the address of the centre nearest you online (european-union.europa.eu/contact-eu/meet-us_en).

On the phone or in writing

Europe Direct is a service that answers your questions about the European Union. You can contact this service:

— by freephone: 00 800 6 7 8 9 10 11 (certain operators may charge for these calls),

— at the following standard number: +32 22999696,

— via the following form: european-union.europa.eu/contact-eu/write-us_en.

Finding information about the EU

Online

Information about the European Union in all the official languages of the EU is available on the Europa website (european-union.europa.eu).

EU publications

You can view or order EU publications at op.europa.eu/en/publications. Multiple copies of free publications can be obtained by contacting Europe Direct or your local documentation centre (european-union.europa.eu/contact-eu/meet-us_en).

EU law and related documents

For access to legal information from the EU, including all EU law since 1951 in all the official language versions, go to EUR-Lex (eur-lex.europa.eu).

EU open data

The portal data.europa.eu provides access to open datasets from the EU institutions, bodies and agencies. These can be downloaded and reused for free, for both commercial and non-commercial purposes. The portal also provides access to a wealth of datasets from European countries.